

```
1  LIBRARY ieee;
2  USE ieee.std_logic_1164.ALL;
3
4  ENTITY code_lock_tester IS
5
6      PORT (
7          --INPUTS
8          CLOCK_50 : IN STD_LOGIC;
9          KEY : IN STD_LOGIC_VECTOR(1 DOWNTO 0);
10         SW : IN STD_LOGIC_VECTOR(3 DOWNTO 0);
11         --OUTPUTS
12         LEDR : OUT STD_LOGIC_VECTOR(9 DOWNTO 0);
13         GPIO : OUT STD_LOGIC_VECTOR(0 DOWNTO 0)
14     );
15 END code_lock_tester;
16
17 ARCHITECTURE stuck OF code_lock_tester IS
18     SIGNAL fails : STD_LOGIC_VECTOR(1 DOWNTO 0);
19     SIGNAL err : STD_LOGIC_VECTOR(0 DOWNTO 0);
20     SIGNAL lock_temp : STD_LOGIC;
21 BEGIN
22     poo_poo : ENTITY work.code_lock_simple PORT MAP
23     (
24         clk => CLOCK_50,
25         reset => KEY(0),
26         enter => KEY(1),
27         code => SW(3 DOWNTO 0),
28         lock => LEDR(0),
29         lock0 => GPIO(0),
30         err_count => LEDR(9 DOWNTO 8)
31     );
32 END stuck;
```